TOSHIBA Bipolar Linear Integrated Circuit Silicon Monolithic

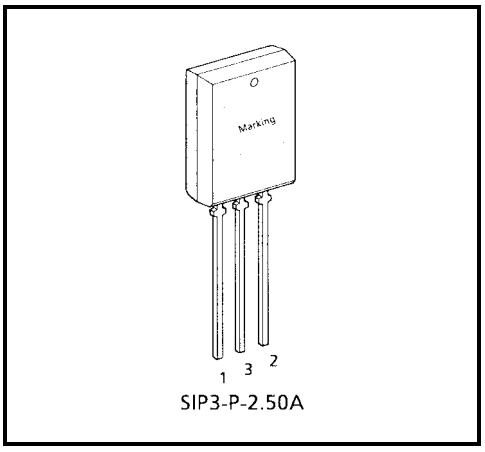
TA79005SB,TA79006SB,TA79007SB,TA79008SB,TA79009SB,TA79010SB, TA79012SB,TA79015SB,TA79018SB,TA79020SB,TA79024SB

Output Current of 1A, Three-Terminal Negative Voltage Regulators

−5 V, −6 V, −7 V, −8 V, −9 V, −10 V, −12 V, −15 V, −18 V, −20 V, −24 V

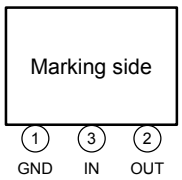
Features

- Suitable for CMOS, TTL, and the other digital IC power supply.
- Internal thermal overload protecting.
- Internal short circuit current limiting.
- Maximum output current of 1.0 A.
- Package in the plastic case TPL (P_D = 1.8 W).

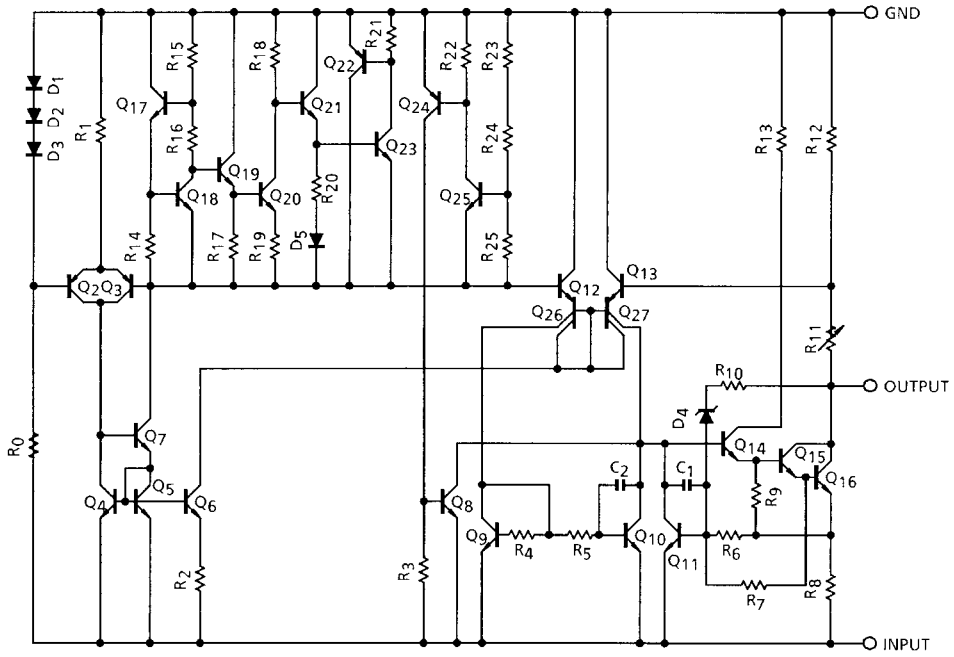


Weight: 1.5 g (Typ.)

Pin Assignment



Equivalent Circuit



Maximum Ratings (Ta = 25°C)

Characteristics		Symbol	Rating	Unit
Input voltage	TA79005SB	V_{IN}	−35	V
	TA79006SB			
	TA79007SB			
	TA79008SB			
	TA79009SB			
	TA79010SB			
	TA79012SB			
	TA79015SB			
	TA79018SB		−40	
	TA79020SB			
	TA79024SB			
Power dissipation	(Ta = 25°C)	P_D	1.8	W
Operating temperature		T_{opr}	−30~85	°C
Storage temperature		T_{stg}	−55~150	°C
Junction temperature		T_j	150	°C
Thermal resistance		$R_{th\ (j-a)}$	69.5	°C/W

TA79005SB
Electrical Characteristics
(Unless otherwise specified, V_{IN} = −10 V, I_{OUT} = 500 mA, 0°C ≤ T_j ≤ 125°C, C_{IN} = 0.33 μF, C_{OUT} = 0.1 μF)

Characteristics		Symbol	Test Circuit	Test Condition		Min	Typ.	Max	Unit
Output voltage		V_{OUT}	1	T_j = 25°C		−5.2	−5.0	−4.8	V
Line regulation		Reg·line	1	T_j = 25°C	−12 V ≤ V_{IN} ≤ −8 V	—	7	50	mV
					−25 V ≤ V_{IN} ≤ −7 V	—	35	100	
Load regulation		Reg·load	1	T_j = 25°C	5 mA ≤ I_{OUT} ≤ 1.5 A	—	11	100	mV
					250 mA ≤ I_{OUT} ≤ 750 mA	—	4	50	
Output voltage		V_{OUT}	1	T_j = 25°C	−20 V ≤ V_{IN} ≤ −7 V, 5 mA ≤ I_{OUT} ≤ 1.0 A	−5.25	—	−4.75	V
Quiescent current		I_B	1	T_j = 25°C		—	4.3	8.0	mA
Quiescent current change	Line	ΔI_B	1	T_j = 25°C	−25 V ≤ V_{IN} ≤ −7 V	—	—	1.3	mA
	Load		1		5 mA ≤ I_{OUT} ≤ 1.0 A	—	—	0.5	
Output noise voltage		V_{NO}	2	Ta = 25°C, I_{OUT} = 20 mA, 10 Hz ≤ f ≤ 100 kHz		—	40	—	μV_{rms}
Ripple rejection		R.R.	3	f = 120 Hz, I_{OUT} = 20 mA, T_j = 25°C		63	70	—	dB
Short circuit current limit		I_{SC}	1	T_j = 25°C		—	1.9	—	A
Dropout voltage		V_D	1	T_j = 25°C, I_{OUT} = 1.0 A		—	2.0	—	V
Average temperature coefficient of output voltage		T_{CVO}	1	I_{OUT} = 5.0 mA		—	0.6	—	mV/°C

TA79006SB
Electrical Characteristics
(Unless otherwise specified, V_{IN} = −11 V, I_{OUT} = 500 mA, 0°C ≤ T_j ≤ 125°C, C_{IN} = 0.33 μF, C_{OUT} = 0.1 μF)

Characteristics		Symbol	Test Circuit	Test Condition		Min	Typ.	Max	Unit
Output voltage		V_{OUT}	1	T_j = 25°C		−6.25	−6.0	−5.75	V
Line regulation		Reg·line	1	T_j = 25°C	−13 V ≤ V_{IN} ≤ −9 V	—	9	60	mV
					−25 V ≤ V_{IN} ≤ −8 V	—	43	120	
Load regulation		Reg·load	1	T_j = 25°C	5 mA ≤ I_{OUT} ≤ 1.5 A	—	13	120	mV
					250 mA ≤ I_{OUT} ≤ 750 mA	—	5	60	
Output voltage		V_{OUT}	1	T_j = 25°C	−21 V ≤ V_{IN} ≤ −8 V, 5 mA ≤ I_{OUT} ≤ 1.0 A	−6.3	—	−5.7	V
Quiescent current		I_B	1	T_j = 25°C		—	4.3	8.0	mA
Quiescent current change	Line	ΔI_B	1	T_j = 25°C	−25 V ≤ V_{IN} ≤ −8 V	—	—	1.3	mA
	Load		1		5 mA ≤ I_{OUT} ≤ 1.0 A	—	—	0.5	
Output noise voltage		V_{NO}	2	Ta = 25°C, I_{OUT} = 20 mA, 10 Hz ≤ f ≤ 100 kHz		—	45	—	μV_{rms}
Ripple rejection		R.R.	3	f = 120 Hz, I_{OUT} = 20 mA, T_j = 25°C		61	68	—	dB
Short circuit current limit		I_{SC}	1	T_j = 25°C		—	1.9	—	A
Dropout voltage		V_D	1	T_j = 25°C, I_{OUT} = 1.0 A		—	2.0	—	V
Average temperature coefficient of output voltage		T_{CVO}	1	I_{OUT} = 5.0 mA		—	0.7	—	mV/°C

TA79007SB
Electrical Characteristics
(Unless otherwise specified, V_{IN} = −12 V, I_{OUT} = 500 mA, 0°C ≤ T_j ≤ 125°C, C_{IN} = 0.33 μF, C_{OUT} = 0.1 μF)

Characteristics		Symbol	Test Circuit	Test Condition		Min	Typ.	Max	Unit
Output voltage		V_{OUT}	1	T_j = 25°C		−7.28	−7.0	−6.72	V
Line regulation		Reg·line	1	T_j = 25°C	−15 V ≤ V_{IN} ≤ −10 V	—	10	70	mV
					−25 V ≤ V_{IN} ≤ −9 V	—	45	140	
Load regulation		Reg·load	1	T_j = 25°C	5 mA ≤ I_{OUT} ≤ 1.5 A	—	20	140	mV
					250 mA ≤ I_{OUT} ≤ 750 mA	—	7	70	
Output voltage		V_{OUT}	1	T_j = 25°C	−22 V ≤ V_{IN} ≤ −9 V, 5 mA ≤ I_{OUT} ≤ 1.0 A	−7.35	—	−6.65	V
Quiescent current		I_B	1	T_j = 25°C		—	4.3	8.0	mA
Quiescent current change	Line	ΔI_B	1	T_j = 25°C	−25 V ≤ V_{IN} ≤ −9 V	—	—	1.0	mA
	Load		1		5 mA ≤ I_{OUT} ≤ 1.0 A	—	—	0.5	
Output noise voltage		V_{NO}	2	Ta = 25°C, I_{OUT} = 20 mA, 10 Hz ≤ f ≤ 100 kHz		—	49	—	μV_{rms}
Ripple rejection		R.R.	3	f = 120 Hz, I_{OUT} = 20 mA, T_j = 25°C		60	67	—	dB
Short circuit current limit		I_{SC}	1	T_j = 25°C		—	1.9	—	A
Dropout voltage		V_D	1	T_j = 25°C, I_{OUT} = 1.0 A		—	2.0	—	V
Average temperature coefficient of output voltage		T_{CVO}	1	I_{OUT} = 5.0 mA		—	0.9	—	mV/°C

TA79008SB
Electrical Characteristics
(Unless otherwise specified, V_{IN} = −14 V, I_{OUT} = 500 mA, 0°C ≤ T_j ≤ 125°C, C_{IN} = 0.33 μF, C_{OUT} = 0.1 μF)

Characteristics		Symbol	Test Circuit	Test Condition		Min	Typ.	Max	Unit
Output voltage		V_{OUT}	1	T_j = 25°C		−8.3	−8.0	−7.7	V
Line regulation		Reg·line	1	T_j = 25°C	−17 V ≤ V_{IN} ≤ −11 V	—	11	80	mV
					−25 V ≤ V_{IN} ≤ −10.5 V	—	47	160	
Load regulation		Reg·load	1	T_j = 25°C	5 mA ≤ I_{OUT} ≤ 1.5 A	—	26	160	mV
					250 mA ≤ I_{OUT} ≤ 750 mA	—	9	80	
Output voltage		V_{OUT}	1	T_j = 25°C	−23 V ≤ V_{IN} ≤ −10.5 V, 5 mA ≤ I_{OUT} ≤ 1.0 A	−8.4	—	−7.6	V
Quiescent current		I_B	1	T_j = 25°C		—	4.3	8.0	mA
Quiescent current change	Line	ΔI_B	1	T_j = 25°C	−25 V ≤ V_{IN} ≤ −10.5 V	—	—	1.0	mA
	Load		1		5 mA ≤ I_{OUT} ≤ 1.0 A	—	—	0.5	
Output noise voltage		V_{NO}	2	Ta = 25°C, I_{OUT} = 20 mA, 10 Hz ≤ f ≤ 100 kHz		—	52	—	μV_{rms}
Ripple rejection		R.R.	3	f = 120 Hz, I_{OUT} = 20 mA, T_j = 25°C		59	66	—	dB
Short circuit current limit		I_{SC}	1	T_j = 25°C		—	1.9	—	A
Dropout voltage		V_D	1	T_j = 25°C, I_{OUT} = 1.0 A		—	2.0	—	V
Average temperature coefficient of output voltage		T_{CVO}	1	I_{OUT} = 5.0 mA		—	1.0	—	mV/°C

TA79009SB
Electrical Characteristics
(Unless otherwise specified, V_{IN} = −15 V, I_{OUT} = 500 mA, 0°C ≤ T_j ≤ 125°C, C_{IN} = 0.33 μF, C_{OUT} = 0.1 μF)

Characteristics		Symbol	Test Circuit	Test Condition		Min	Typ.	Max	Unit
Output voltage		V_{OUT}	1	T_j = 25°C		−9.3	−9.0	−8.7	V
Line regulation		Reg·line	1	T_j = 25°C	−19 V ≤ V_{IN} ≤ −13 V	—	11	82	mV
					−26 V ≤ V_{IN} ≤ −11.5 V	—	48	162	
Load regulation		Reg·load	1	T_j = 25°C	5 mA ≤ I_{OUT} ≤ 1.5 A	—	33	162	mV
					250 mA ≤ I_{OUT} ≤ 750 mA	—	11	82	
Output voltage		V_{OUT}	1	T_j = 25°C	−24 V ≤ V_{IN} ≤ −11.5 V, 5 mA ≤ I_{OUT} ≤ 1.0 A	−9.4	—	−8.6	V
Quiescent current		I_B	1	T_j = 25°C		—	4.3	8.0	mA
Quiescent current change	Line	ΔI_B	1	T_j = 25°C	−26.5 V ≤ V_{IN} ≤ −13 V	—	—	1.0	mA
	Load		1		5 mA ≤ I_{OUT} ≤ 1.0 A	—	—	0.5	
Output noise voltage		V_{NO}	2	Ta = 25°C, I_{OUT} = 20 mA, 10 Hz ≤ f ≤ 100 kHz		—	60	—	μV_{rms}
Ripple rejection		R.R.	3	f = 120 Hz, I_{OUT} = 20 mA, T_j = 25°C		57	64	—	dB
Short circuit current limit		I_{SC}	1	T_j = 25°C		—	1.9	—	A
Dropout voltage		V_D	1	T_j = 25°C, I_{OUT} = 1.0 A		—	2.0	—	V
Average temperature coefficient of output voltage		T_{CVO}	1	I_{OUT} = 5.0 mA		—	1.1	—	mV/°C

TA79010SB
Electrical Characteristics
(Unless otherwise specified, V_{IN} = −16 V, I_{OUT} = 500 mA, 0°C ≤ T_j ≤ 125°C, C_{IN} = 0.33 μF, C_{OUT} = 0.1 μF)

Characteristics		Symbol	Test Circuit	Test Condition		Min	Typ.	Max	Unit
Output voltage		V_{OUT}	1	T_j = 25°C		−10.4	−10.0	−9.6	V
Line regulation		Reg·line	1	T_j = 25°C	−20 V ≤ V_{IN} ≤ −14 V	—	12	90	mV
					−27 V ≤ V_{IN} ≤ −12.5 V	—	50	180	
Load regulation		Reg·load	1	T_j = 25°C	5 mA ≤ I_{OUT} ≤ 1.5 A	—	40	180	mV
					250 mA ≤ I_{OUT} ≤ 750 mA	—	13	90	
Output voltage		V_{OUT}	1	T_j = 25°C	−25 V ≤ V_{IN} ≤ −12.5 V, 5 mA ≤ I_{OUT} ≤ 1.0 A	−10.5	—	−9.5	V
Quiescent current		I_B	1	T_j = 25°C		—	4.4	8.0	mA
Quiescent current change	Line	ΔI_B	1	T_j = 25°C	−27.5 V ≤ V_{IN} ≤ −14 V	—	—	1.0	mA
	Load		1		5 mA ≤ I_{OUT} ≤ 1.0 A	—	—	0.5	
Output noise voltage		V_{NO}	2	Ta = 25°C, I_{OUT} = 20 mA, 10 Hz ≤ f ≤ 100 kHz		—	65	—	μV_{rms}
Ripple rejection		R.R.	3	f = 120 Hz, I_{OUT} = 20 mA, T_j = 25°C		57	63	—	dB
Short circuit current limit		I_{SC}	1	T_j = 25°C		—	1.9	—	A
Dropout voltage		V_D	1	T_j = 25°C, I_{OUT} = 1.0 A		—	2.0	—	V
Average temperature coefficient of output voltage		T_{CVO}	1	I_{OUT} = 5.0 mA		—	1.3	—	mV/°C

TA79012SB
Electrical Characteristics
(Unless otherwise specified, V_{IN} = −19 V, I_{OUT} = 500 mA, 0°C ≤ T_j ≤ 125°C, C_{IN} = 0.33 μF, C_{OUT} = 0.1 μF)

Characteristics		Symbol	Test Circuit	Test Condition		Min	Typ.	Max	Unit
Output voltage		V_{OUT}	1	T_j = 25°C		−12.5	−12.0	−11.5	V
Line regulation		Reg·line	1	T_j = 25°C	−22 V ≤ V_{IN} ≤ −16 V	—	13	120	mV
					−30 V ≤ V_{IN} ≤ −14.5 V	—	55	240	
Load regulation		Reg·load	1	T_j = 25°C	5 mA ≤ I_{OUT} ≤ 1.5 A	—	46	240	mV
					250 mA ≤ I_{OUT} ≤ 750 mA	—	17	120	
Output voltage		V_{OUT}	1	T_j = 25°C	−27 V ≤ V_{IN} ≤ −14.5 V, 5 mA ≤ I_{OUT} ≤ 1.0 A	−12.6	—	−11.4	V
Quiescent current		I_B	1	T_j = 25°C		—	4.4	8.0	mA
Quiescent current change	Line	ΔI_B	1	T_j = 25°C	−30 V ≤ V_{IN} ≤ −14.5 V	—	—	1.0	mA
	Load		1		5 mA ≤ I_{OUT} ≤ 1.0 A	—	—	0.5	
Output noise voltage		V_{NO}	2	Ta = 25°C, I_{OUT} = 20 mA, 10 Hz ≤ f ≤ 100 kHz		—	75	—	μV_{rms}
Ripple rejection		R.R.	3	f = 120 Hz, I_{OUT} = 20 mA, T_j = 25°C		54	61	—	dB
Short circuit current limit		I_{SC}	1	T_j = 25°C		—	1.9	—	A
Dropout voltage		V_D	1	T_j = 25°C, I_{OUT} = 1.0 A		—	2.0	—	V
Average temperature coefficient of output voltage		T_{CVO}	1	I_{OUT} = 5.0 mA		—	1.6	—	mV/°C

TA79015SB
Electrical Characteristics
(Unless otherwise specified, V_{IN} = −23 V, I_{OUT} = 500 mA, 0°C ≤ T_j ≤ 125°C, C_{IN} = 0.33 μF, C_{OUT} = 0.1 μF)

Characteristics		Symbol	Test Circuit	Test Condition		Min	Typ.	Max	Unit
Output voltage		V_{OUT}	1	T_j = 25°C		−15.6	−15.0	−14.4	V
Line regulation		Reg·line	1	T_j = 25°C	−26 V ≤ V_{IN} ≤ −20 V	—	14	150	mV
					−30 V ≤ V_{IN} ≤ −17.5 V	—	57	300	
Load regulation		Reg·load	1	T_j = 25°C	5 mA ≤ I_{OUT} ≤ 1.5 A	—	68	300	mV
					250 mA ≤ I_{OUT} ≤ 750 mA	—	25	150	
Output voltage		V_{OUT}	1	T_j = 25°C	−30 V ≤ V_{IN} ≤ −17.5 V, 5 mA ≤ I_{OUT} ≤ 1.0 A	−15.75	—	−14.25	V
Quiescent current		I_B	1	T_j = 25°C		—	4.4	8.0	mA
Quiescent current change	Line	ΔI_B	1	T_j = 25°C	−30 V ≤ V_{IN} ≤ −17.5 V	—	—	1.0	mA
	Load		1		5 mA ≤ I_{OUT} ≤ 1.0 A	—	—	0.5	
Output noise voltage		V_{NO}	2	Ta = 25°C, I_{OUT} = 20 mA, 10 Hz ≤ f ≤ 100 kHz		—	90	—	μV_{rms}
Ripple rejection		R.R.	3	f = 120 Hz, I_{OUT} = 20 mA, T_j = 25°C		53	60	—	dB
Short circuit current limit		I_{SC}	1	T_j = 25°C		—	1.9	—	A
Dropout voltage		V_D	1	T_j = 25°C, I_{OUT} = 1.0 A		—	2.0	—	V
Average temperature coefficient of output voltage		T_{CVO}	1	I_{OUT} = 5.0 mA		—	2.0	—	mV/°C

TA79018SB
Electrical Characteristics
(Unless otherwise specified, V_{IN} = −27 V, I_{OUT} = 500 mA, 0°C ≤ T_j ≤ 125°C, C_{IN} = 0.33 μF, C_{OUT} = 0.1 μF)

Characteristics		Symbol	Test Circuit	Test Condition		Min	Typ.	Max	Unit
Output voltage		V_{OUT}	1	T_j = 25°C		−18.7	−18.0	−17.3	V
Line regulation		Reg·line	1	T_j = 25°C	−30 V ≤ V_{IN} ≤ −24 V	—	25	180	mV
					−33 V ≤ V_{IN} ≤ −21 V	—	80	360	
Load regulation		Reg·load	1	T_j = 25°C	5 mA ≤ I_{OUT} ≤ 1.5 A	—	110	360	mV
					250 mA ≤ I_{OUT} ≤ 750 mA	—	55	180	
Output voltage		V_{OUT}	1	T_j = 25°C	−33 V ≤ V_{IN} ≤ −21 V, 5 mA ≤ I_{OUT} ≤ 1.0 A	−18.85	—	−17.15	V
Quiescent current		I_B	1	T_j = 25°C		—	4.5	8.0	mA
Quiescent current change	Line	ΔI_B	1	T_j = 25°C	−33 V ≤ V_{IN} ≤ −21 V	—	—	1.0	mA
	Load		1		5 mA ≤ I_{OUT} ≤ 1.0 A	—	—	0.5	
Output noise voltage		V_{NO}	2	Ta = 25°C, I_{OUT} = 20 mA, 10 Hz ≤ f ≤ 100 kHz		—	110	—	μV_{rms}
Ripple rejection		R.R.	3	f = 120 Hz, I_{OUT} = 20 mA, T_j = 25°C		52	59	—	dB
Short circuit current limit		I_{SC}	1	T_j = 25°C		—	1.9	—	A
Dropout voltage		V_D	1	T_j = 25°C, I_{OUT} = 1.0 A		—	2.0	—	V
Average temperature coefficient of output voltage		T_{CVO}	1	I_{OUT} = 5.0 mA		—	2.5	—	mV/°C

TA79020SB
Electrical Characteristics
(Unless otherwise specified, V_{IN} = −30 V, I_{OUT} = 500 mA, 0°C ≤ T_j ≤ 125°C, C_{IN} = 0.33 μF, C_{OUT} = 0.1 μF)

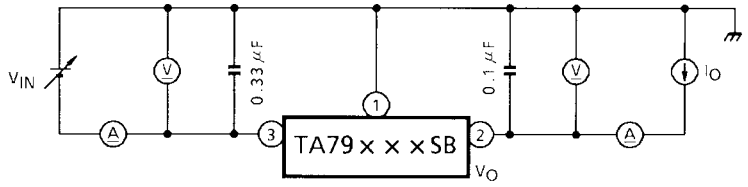
Characteristics		Symbol	Test Circuit	Test Condition		Min	Typ.	Max	Unit
Output voltage		V_{OUT}	1	T_j = 25°C		−20.8	−20.0	−19.2	V
Line regulation		Reg·line	1	T_j = 25°C	−32 V ≤ V_{IN} ≤ −26 V	—	28	180	mV
					−35 V ≤ V_{IN} ≤ −24 V	—	104	360	
Load regulation		Reg·load	1	T_j = 25°C	5 mA ≤ I_{OUT} ≤ 1.5 A	—	130	360	mV
					250 mA ≤ I_{OUT} ≤ 750 mA	—	70	180	
Output voltage		V_{OUT}	1	T_j = 25°C	−35 V ≤ V_{IN} ≤ −24 V, 5 mA ≤ I_{OUT} ≤ 1.0 A	−21.0	—	−19.0	V
Quiescent current		I_B	1	T_j = 25°C		—	4.6	8.0	mA
Quiescent current change	Line	ΔI_B	1	T_j = 25°C	−36.5 V ≤ V_{IN} ≤ −25 V	—	—	1.0	mA
	Load		1		5 mA ≤ I_{OUT} ≤ 1.0 A	—	—	0.5	
Output noise voltage		V_{NO}	2	Ta = 25°C, I_{OUT} = 20 mA, 10 Hz ≤ f ≤ 100 kHz		—	140	—	μV_{rms}
Ripple rejection		R.R.	3	f = 120 Hz, I_{OUT} = 20 mA, T_j = 25°C		50	57	—	dB
Short circuit current limit		I_{SC}	1	T_j = 25°C		—	1.9	—	A
Dropout voltage		V_D	1	T_j = 25°C, I_{OUT} = 1.0 A		—	2.0	—	V
Average temperature coefficient of output voltage		T_{CVO}	1	I_{OUT} = 5.0 mA		—	3.0	—	mV/°C

TA79024SB
Electrical Characteristics
(Unless otherwise specified, V_{IN} = −33 V, I_{OUT} = 500 mA, 0°C ≤ T_j ≤ 125°C, C_{IN} = 0.33 μF, C_{OUT} = 0.1 μF)

Characteristics		Symbol	Test Circuit	Test Condition		Min	Typ.	Max	Unit
Output voltage		V_{OUT}	1	T_j = 25°C		−25.0	−24.0	−23.0	V
Line regulation		Reg·line	1	T_j = 25°C	−36 V ≤ V_{IN} ≤ −30 V	—	31	240	mV
					−38 V ≤ V_{IN} ≤ −27 V	—	118	480	
Load regulation		Reg·load	1	T_j = 25°C	5 mA ≤ I_{OUT} ≤ 1.5 A	—	150	480	mV
					250 mA ≤ I_{OUT} ≤ 750 mA	—	85	240	
Output voltage		V_{OUT}	1	T_j = 25°C	−38 V ≤ V_{IN} ≤ −27 V, 5 mA ≤ I_{OUT} ≤ 1.0 A	−25.2	—	−22.8	V
Quiescent current		I_B	1	T_j = 25°C		—	4.6	8.0	mA
Quiescent current change	Line	ΔI_B	1	T_j = 25°C	−38 V ≤ V_{IN} ≤ −27 V	—	—	1.0	mA
	Load		1		5 mA ≤ I_{OUT} ≤ 1.0 A	—	—	0.5	
Output noise voltage		V_{NO}	2	Ta = 25°C, I_{OUT} = 20 mA, 10 Hz ≤ f ≤ 100 kHz		—	170	—	μV_{rms}
Ripple rejection		R.R.	3	f = 120 Hz, I_{OUT} = 20 mA, T_j = 25°C		49	56	—	dB
Short circuit current limit		I_{SC}	1	T_j = 25°C		—	1.9	—	A
Dropout voltage		V_D	1	T_j = 25°C, I_{OUT} = 1.0 A		—	2.0	—	V
Average temperature coefficient of output voltage		T_{CVO}	1	I_{OUT} = 5.0 mA		—	3.5	—	mV/°C

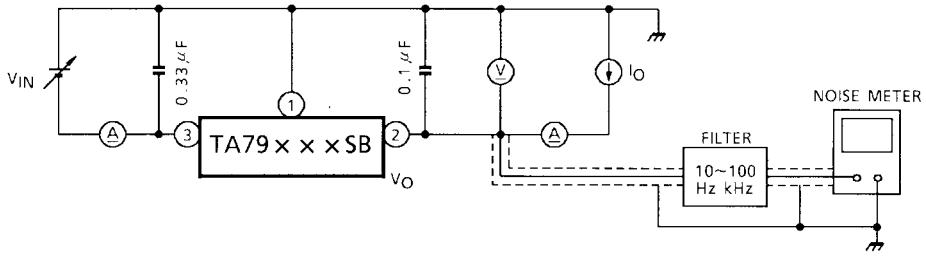
Test Circuit 1

V_{OUT}, Reg·line, Reg·load, I_B, ΔI_B, V_D, T_{CVO}



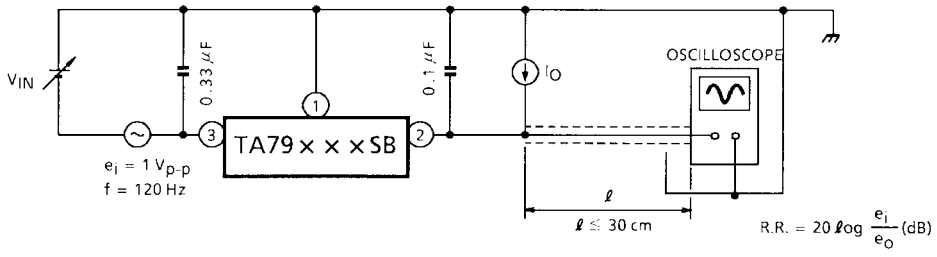
Test Circuit 2

V_{NO}

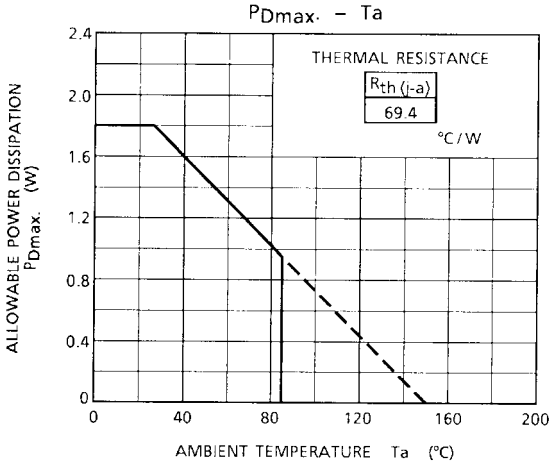


Test Circuit 3

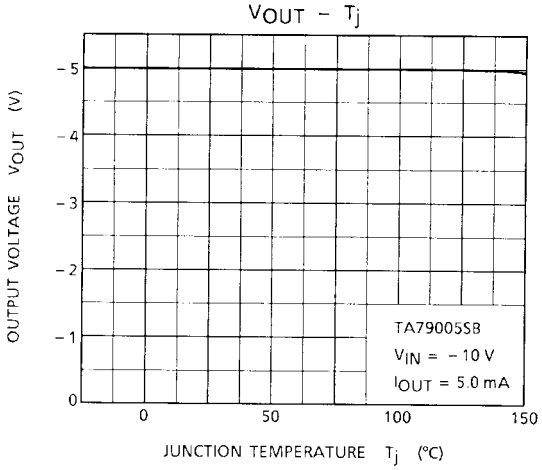
R.R.



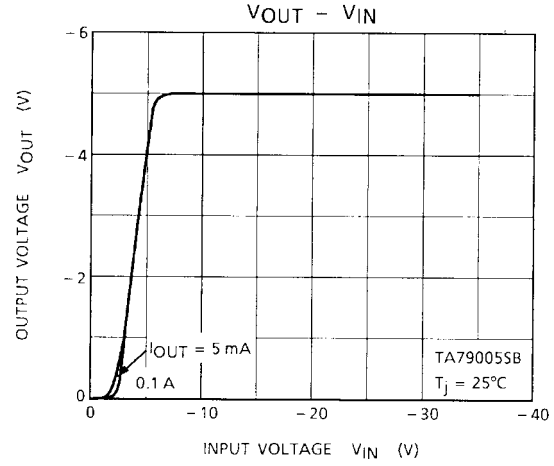
$P_{Dmax.}$ – Ta



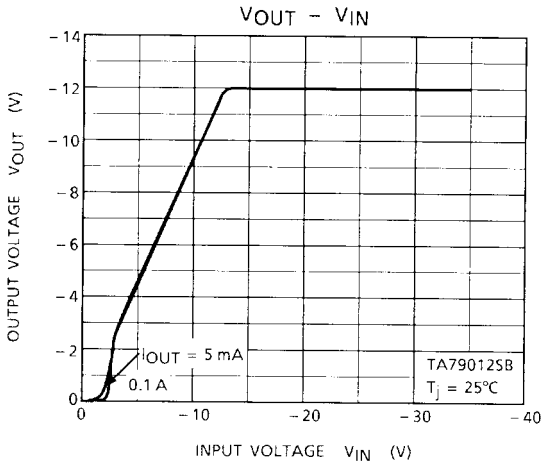
V_{OUT} – T_j



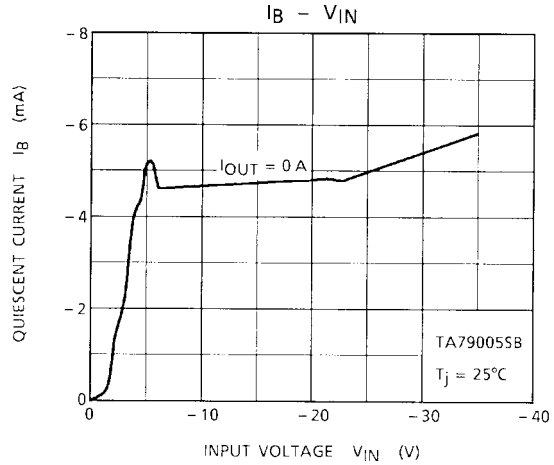
V_{OUT} – V_{IN}



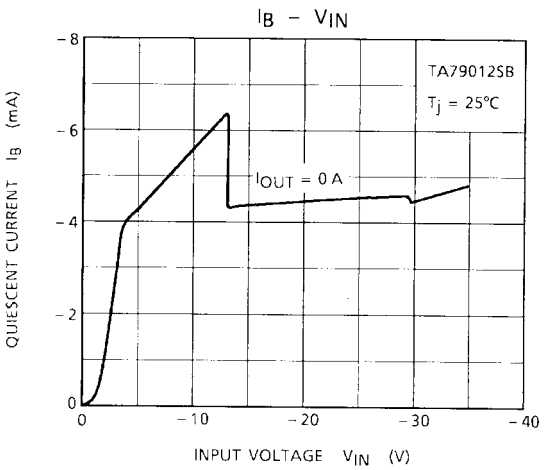
V_{OUT} – V_{IN}



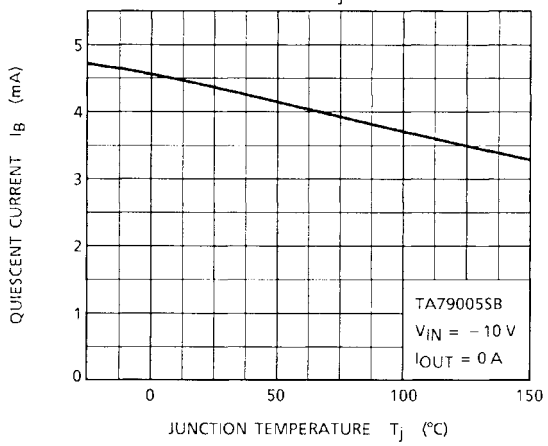
I_B – V_{IN}



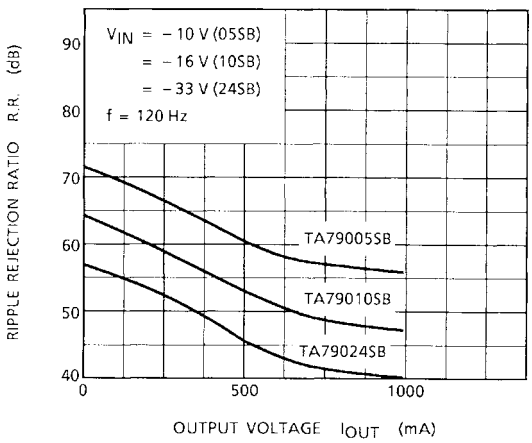
I_B – V_{IN}



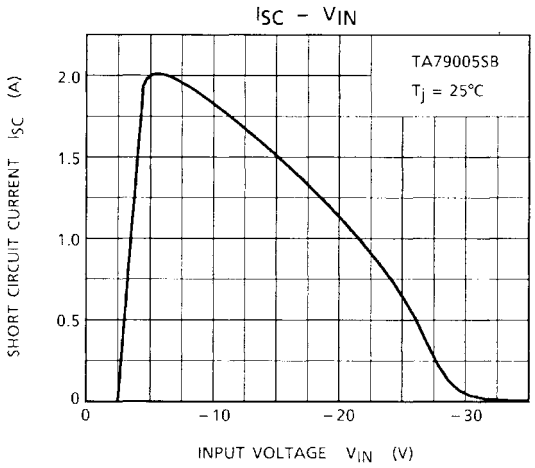
$I_B - T_j$



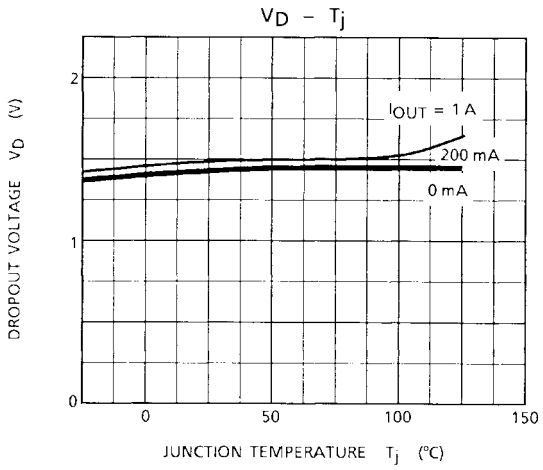
R.R. − I_{OUT}



$I_{SC} - V_{IN}$

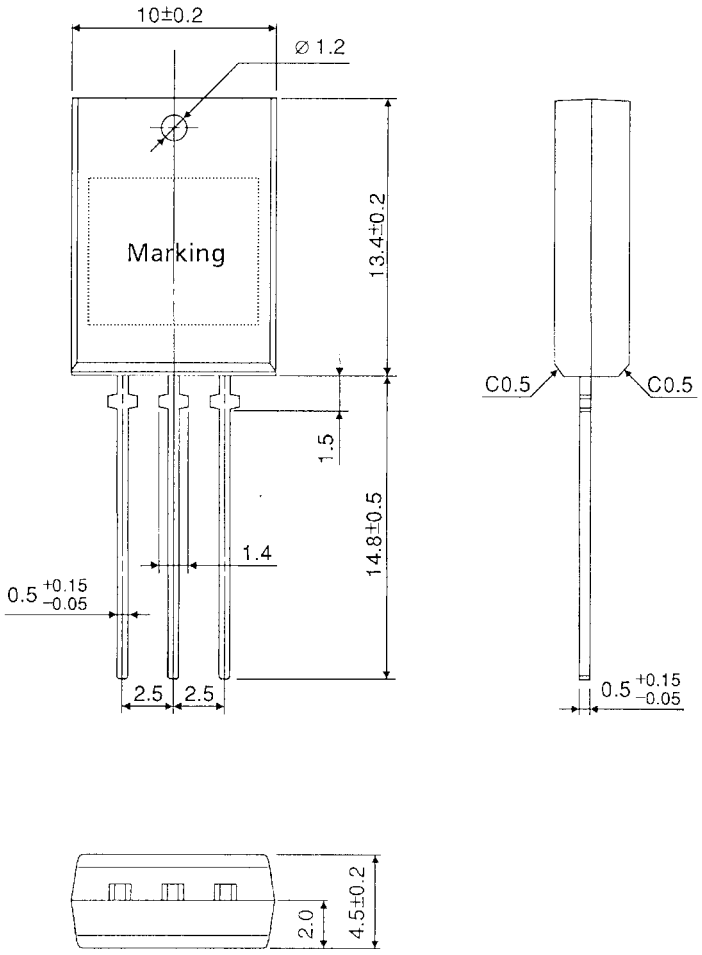


$V_D - T_j$



Package Dimensions

SIP3-P-2.50A Unit : mm



Weight : 1.5 g (Typ.)

RESTRICTIONS ON PRODUCT USE

000707EBA

- TOSHIBA is continually working to improve the quality and reliability of its products. Nevertheless, semiconductor devices in general can malfunction or fail due to their inherent electrical sensitivity and vulnerability to physical stress. It is the responsibility of the buyer, when utilizing TOSHIBA products, to comply with the standards of safety in making a safe design for the entire system, and to avoid situations in which a malfunction or failure of such TOSHIBA products could cause loss of human life, bodily injury or damage to property.
  In developing your designs, please ensure that TOSHIBA products are used within specified operating ranges as set forth in the most recent TOSHIBA products specifications. Also, please keep in mind the precautions and conditions set forth in the “Handling Guide for Semiconductor Devices,” or “TOSHIBA Semiconductor Reliability Handbook” etc..
- The TOSHIBA products listed in this document are intended for usage in general electronics applications (computer, personal equipment, office equipment, measuring equipment, industrial robotics, domestic appliances, etc.). These TOSHIBA products are neither intended nor warranted for usage in equipment that requires extraordinarily high quality and/or reliability or a malfunction or failure of which may cause loss of human life or bodily injury (“Unintended Usage”). Unintended Usage include atomic energy control instruments, airplane or spaceship instruments, transportation instruments, traffic signal instruments, combustion control instruments, medical instruments, all types of safety devices, etc.. Unintended Usage of TOSHIBA products listed in this document shall be made at the customer’s own risk.
- The products described in this document are subject to the foreign exchange and foreign trade laws.
- The information contained herein is presented only as a guide for the applications of our products. No responsibility is assumed by TOSHIBA CORPORATION for any infringements of intellectual property or other rights of the third parties which may result from its use. No license is granted by implication or otherwise under any intellectual property or other rights of TOSHIBA CORPORATION or others.
- The information contained herein is subject to change without notice.

This datasheet has been download from:

www.datasheetcatalog.com

Datasheets for electronics components.